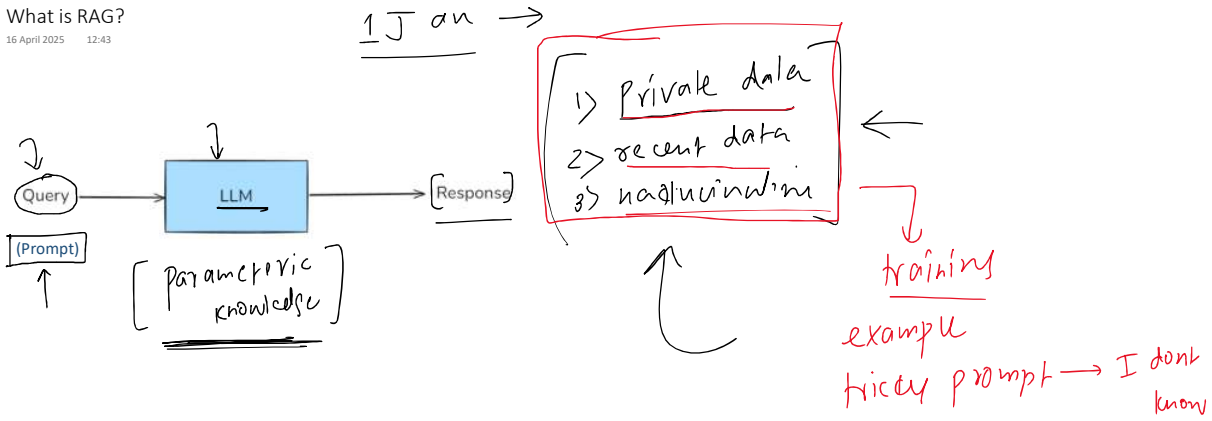
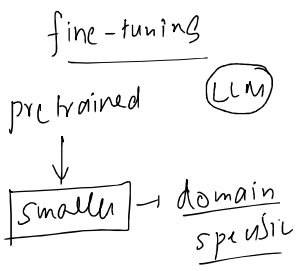
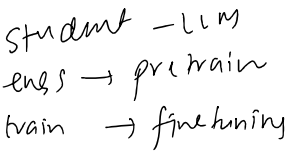
What is RAG?

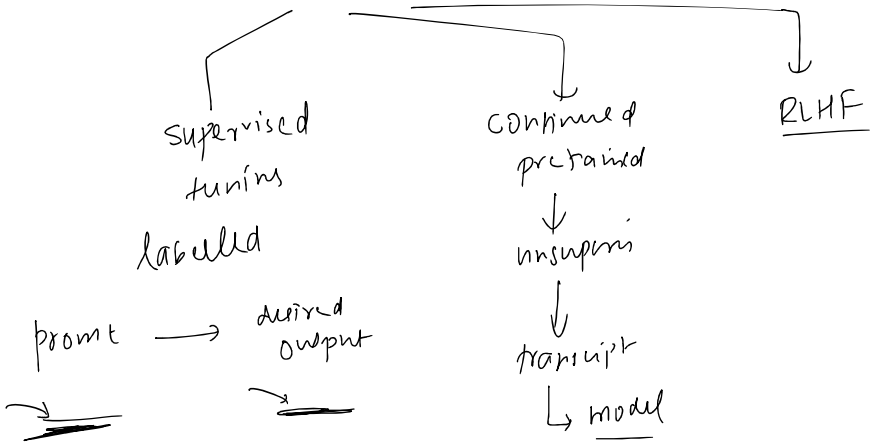
16 April 2025 12:43





1. Collect data	A few hundred – few hundred-thousand carefully curated examples (prompts → desired outputs).
2. Choose a method	Full-parameter FT, LoRA/QLoRA, or parameter-efficient adapters.
3. Train for a few epochs	You keep the base weights frozen or partially frozen and update only a small subset (LoRA) or all weights (full FT).
4. Evaluate & safety-test	Measure exact-match, factuality, and hallucination rate against held-out data; red-team for safety.





1) LLM → train
2) technical expertise
3)

In context learning

In-Context Learning is a core capability of Large Language Models (LLMs) like GPT-3/4, Claude, and Llama, where the model learns to solve a task purely by seeing examples in the prompt—without updating its weights.

LLM → emergent property

```
Below are examples of texts labeled with their sentiment.
Use these examples to determine the sentiment of the final text.

Text: I love this phone. It's so smooth. → Positive
Text: This app crashes a lot. → Negative
Text: The camera is amazing! → Positive
```

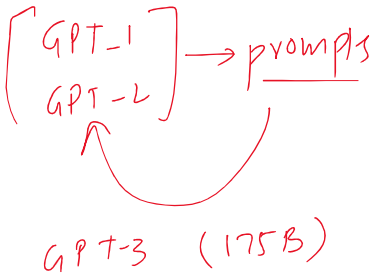
LLM → emergent property

```
Use these examples to determine the sentiment of the final text.

Text: I love this phone. It's so smooth. → Positive
Text: This app crashes a lot. → Negative
Text: The camera is amazing! → Positive

Text: I hate the battery life. →
```

An **emergent property** is a behaviour or ability that suddenly appears in a system when it reaches a certain scale or complexity—even though it was not explicitly programmed or expected from the individual components.

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Label the named entities in the sentences (Person, Location, Organization):

Sentence: Elon Musk founded SpaceX.
Entities: [Elon Musk → Person], [SpaceX → Organization]

Sentence: The Eiffel Tower is in Paris.
Entities: [Eiffel Tower → Location], [Paris → Location]

Sentence: Sundar Pichai leads Google.
Entities:
```

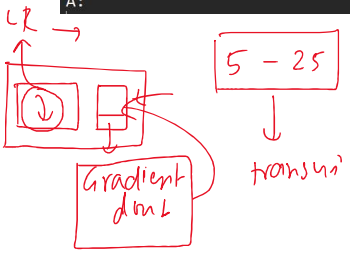
```
Solve the math problems step by step:

Q: John has 3 apples. He buys 4 more. How many apples does he have now?
A: John had 3 apples. He bought 4 more. 3 + 4 = 7. → 7

Q: Sarah has 10 pens. She gives away 3. How many does she have left?
A: Sarah had 10 pens. She gave away 3. 10 - 3 = 7. → 7

Q: A pizza is cut into 8 slices. Alex eats 3 slices. How many are left?
A:
```

context

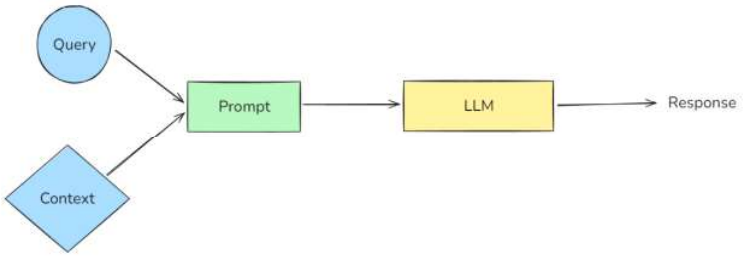


- Instead of just *example tasks*, retrieve **background information, facts, documents, product manuals,** etc.
- Inject that into the prompt to **augment the model's knowledge**

```
"""You are a helpful assistant.
Answer the question ONLY from the provided context.
If the context is insufficient, just say you don't know.

{context}
Question: {question}"""
```

RAG is a way to make a language model (like ChatGPT) smarter by giving it extra information at the time you ask your question.



Understanding RAG

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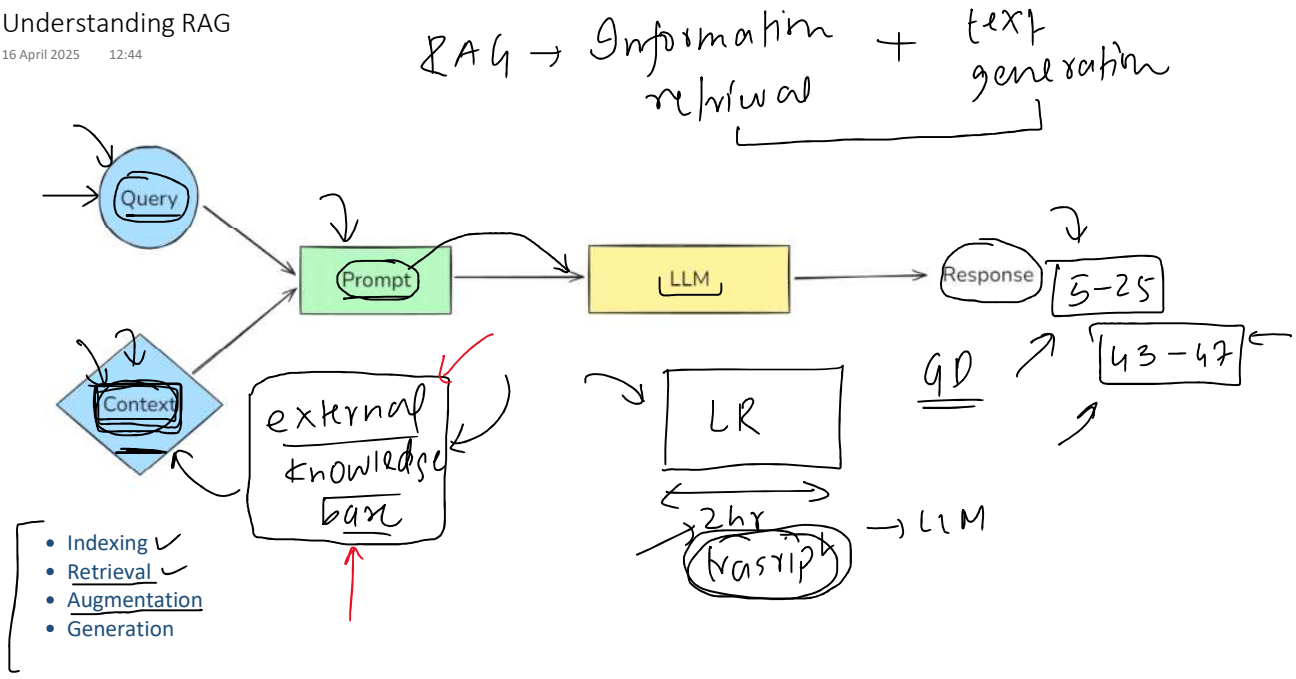


Indexing - **Indexing** is the process of **preparing your knowledge base so that it can be efficiently searched** at query time. This steps consists of 4 sub-steps.

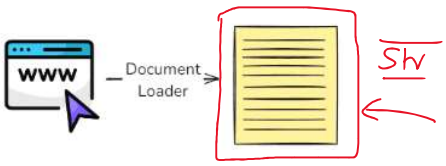
1. Document Ingestion - You load your source knowledge into memory.

   Examples:

   - PDF reports, Word documents
   - YouTube transcripts, blog pages
   - GitHub repos, internal wikis
   - SQL records, scraped webpages

   Tools:

   - LangChain loaders (`PyPDFLoader`, `YoutubeLoader`, `WebBaseLoader`, `GitLoader`, etc.)



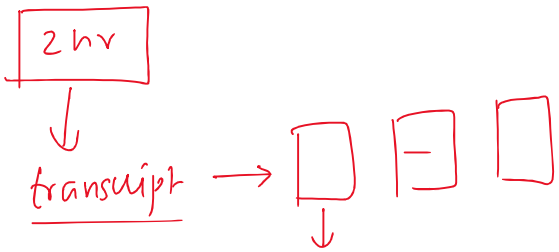
2. Text Chunking - Break large documents into small, semantically meaningful **chunks**

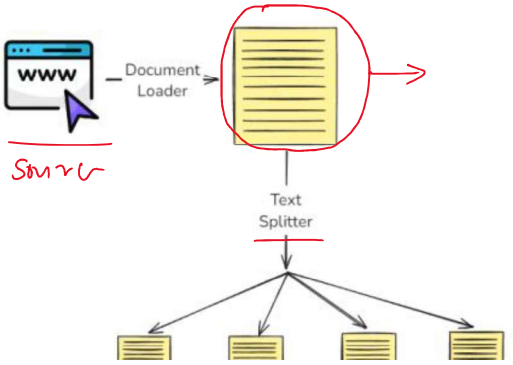
   Why chunk?

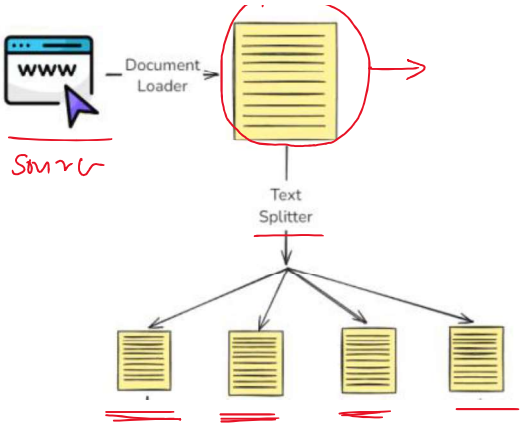
   - LLMs have context limits (e.g., 4K–32K tokens)
   - Smaller chunks are more focused → better semantic search

   Tools:

   - `RecursiveCharacterTextSplitter`, `MarkdownHeaderTextSplitter`, `SemanticChunker`







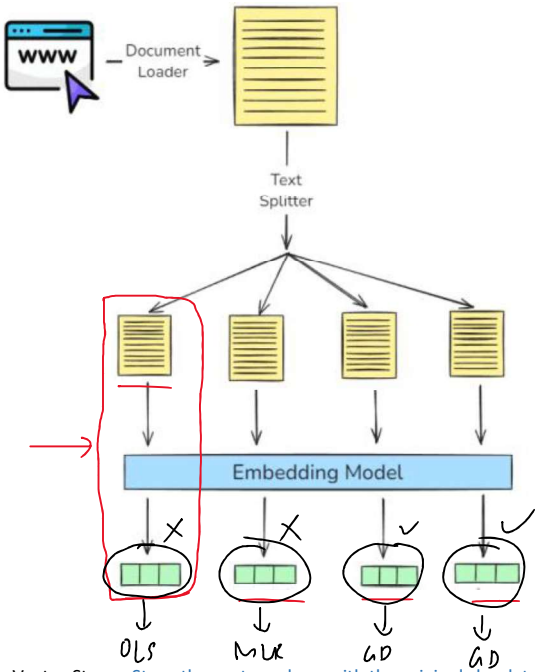
3. Embedding Generation - Convert each chunk into a **dense vector** (embedding) that captures its meaning.

   🔍 Why embeddings?

   - Similar ideas land close together in vector space
   - Allows fast, fuzzy semantic search

   🔧 Tools:

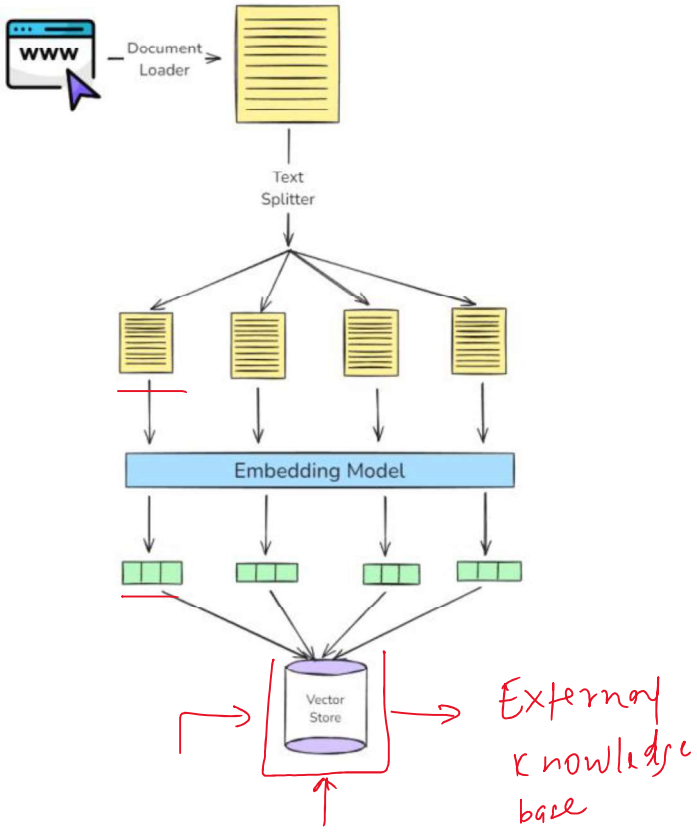
   - `OpenAIEmbeddings`, `SentenceTransformerEmbeddings`, `InstructorEmbeddings`, etc.



4. Storage in a Vector Store - Store the vectors along with the original chunk text + metadata in a **vector database**.

   🔧 Vector DB options:

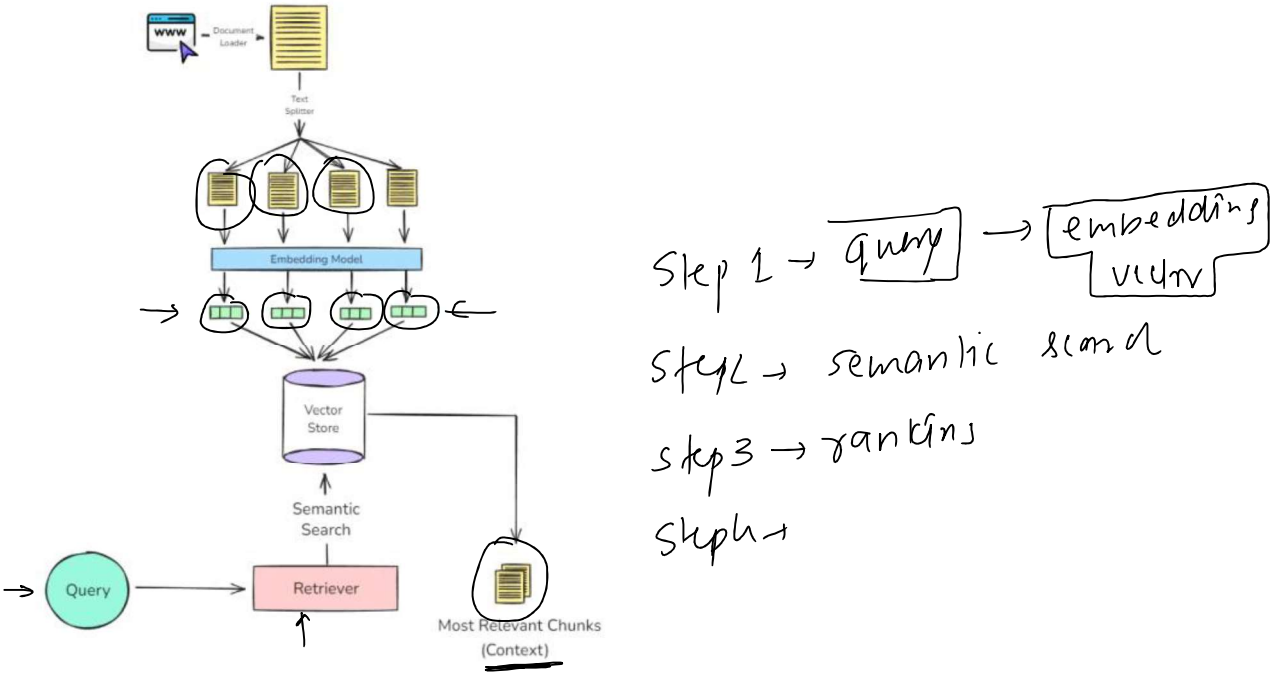
   - Local: `FAISS`, `Chroma`
   - Cloud: `Pinecone`, `Weaviate`, `Milvus`, `Qdrant`



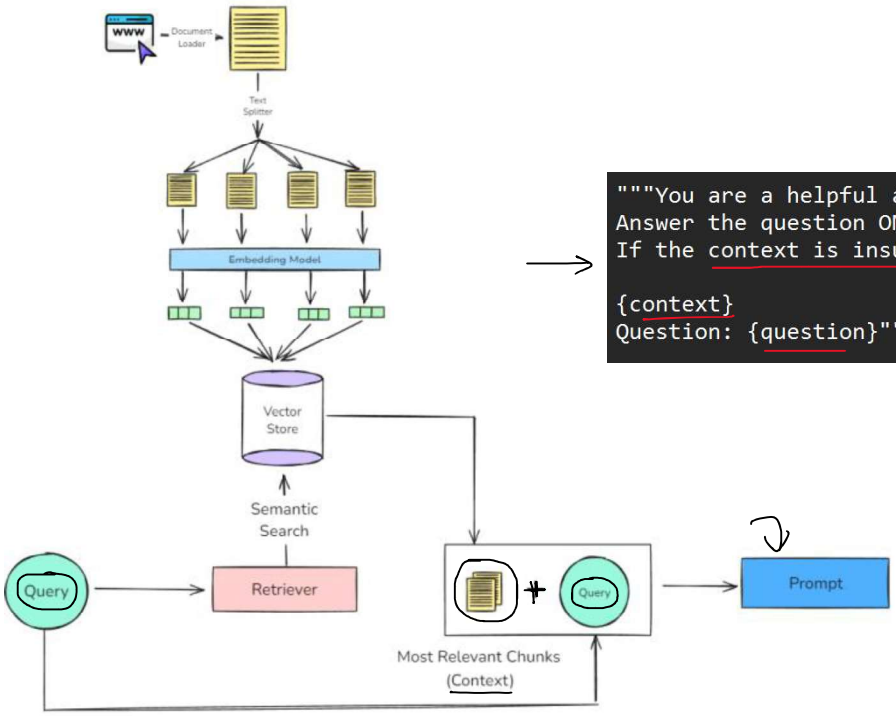
Retrieval - **Retrieval** is the *real-time* process of **finding the most relevant pieces of information** from a pre-built index (created during indexing) based on the user's question.

It's like asking:

> "From all the knowledge I have, which 3–5 chunks are most helpful to answer this query?"



Augmentation - **Augmentation** refers to the step where the **retrieved documents** (chunks of relevant context) are **combined with the user's query** to form a new, enriched prompt for the LLM.




```
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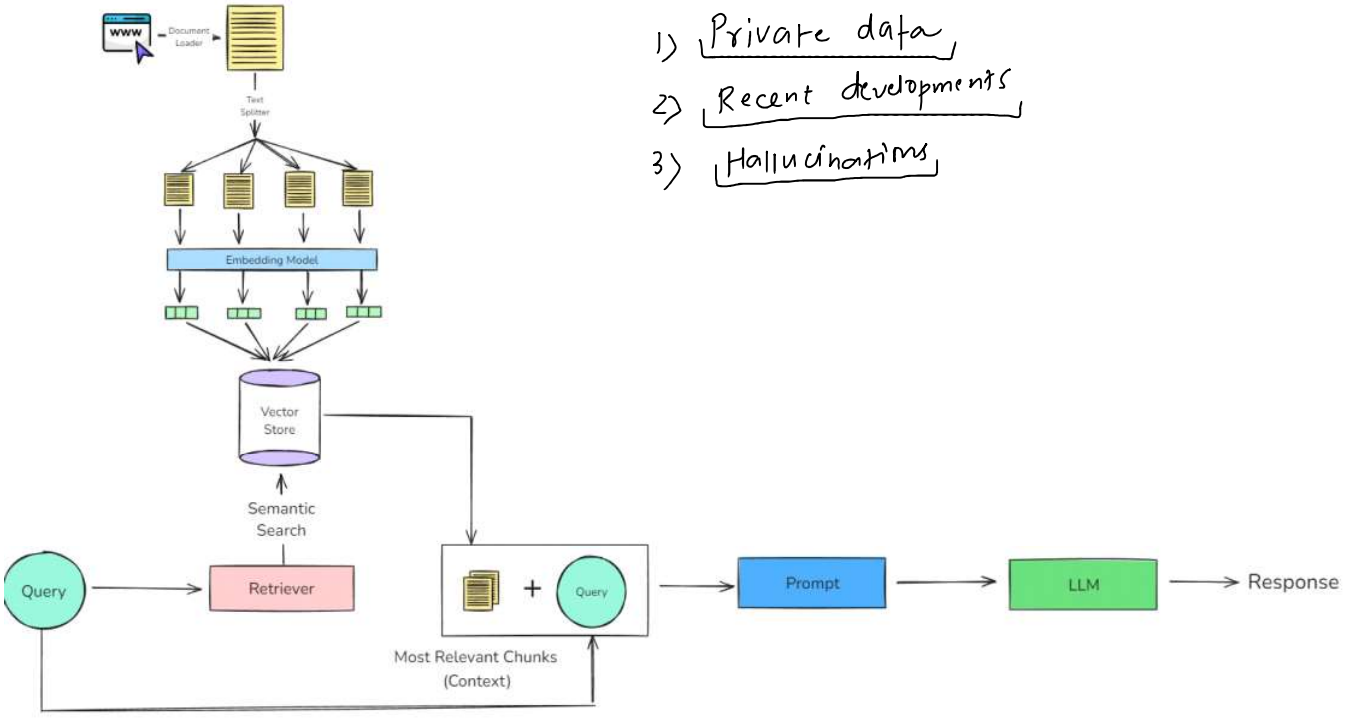
Question: {question}"""
```

Generation - **Generation** is the final step where a **Large Language Model (LLM)** uses the **user's query** and the **retrieved & augmented context** to generate a response.



1) Private data
2) Recent developments
3) Hallucinations